

HASHAN KAVINGA WEERASOORIYA

PhD Candidate · Computational Imaging, Machine Learning, and Single-Photon LiDAR

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EDUCATION

Purdue University, Graduate Student Intelligent Imaging Lab, Electrical & Computer Engineering	GPA: 3.96/4.00	2022–May 2027 (Expected)
BSc of Engineering, University of Peradeniya, Sri Lanka. Electrical and Electronic Engineering	GPA: 3.95/4.00	2015–2020

RESEARCH, PUBLICATIONS, SELECTED PROJECTS

Single-Photon LiDAR Sensing, Simulation, Compression and Reconstruction	2023–Present
📄 Joint Depth and Reflectivity Estimation using Single-Photon LiDAR , <i>Optics Express</i> , 2026 (Under Review). SPLiDER: recovers depth and reflectivity jointly from raw photon timestamps in fast-moving scenes by exploiting their mutual correlation. Skills: Deep Learning, PyTorch, Statistical Modeling, Inverse Problems, Simulation	
📄 SPARC: Scene-Guided Photon Acquisition and Sparse Representation for SP-LiDAR Compression , <i>IEEE WACV</i> , 2026 (Under Review). End-to-end spatio-temporal compression on NVIDIA Jetson Xavier, combining tracking and RGB-guided adaptive encoding with a lightweight neural decoder that improves 3D reconstruction accuracy. Skills: Embedded Systems, Sensor Fusion, Compression Techniques	
📄 Single-Photon LiDAR Lab-Bench Construction , <i>Hardware Project</i> , 2025. Designed and implemented a SP-LiDAR lab-bench integrating TCSPC electronics, single-photon detectors, and galvanometric scanning optics. Skills: Optical System Design, Photon Detection, Data Acquisition	
📄 Resolution Limit of Single-Photon LiDAR , <i>IEEE/CVF CVPR</i> , 2024. Derived the fundamental trade-off between detector-array resolution and per-pixel SNR via the MSE of the ML time-delay estimator. Skills: Signal Processing, Theoretical Analysis	
📄 Fast and Efficient Light Scattering Simulator for SP-LiDAR , <i>Ongoing Project</i> . Developing a fast light scattering simulator to efficiently model photon propagation through scattering media. Skills: Novel View Synthesis, Simulation, Computational Optics	
📄 Real-Time Markov Modeling for Single-Photon LiDAR: 1000× Acceleration and Convergence Analysis , <i>ICASSP</i> , 2026. First non-sequential Markov model of SP-LiDAR timestamps under dead time: 1000× faster stationary-distribution computation, plus convergence analysis. Skills: Probabilistic Modeling, Markov Processes, Algorithm Acceleration	
📄 Markov-Renewal Single-Photon LiDAR Simulator , <i>ECCV</i> , 2026. Physics-consistent Markov-renewal simulation framework for SP-LiDAR, modeling photon arrivals, dead-time effects, and temporal correlations. Skills: Stochastic Processes, Photon Detection Modeling, Simulation	
Hyperspectral Imaging for Remote Sensing and Applications	2019–2022
📄 Constrained NMF for Blind Hyperspectral Unmixing Incorporating Endmember Independence , <i>IEEE JSTARS</i> , 2021. Developed a blind non-negative matrix factorization model enforcing endmember independence achieving superior spectral separation accuracy. Skills: Optimization, Linear Algebra	
📄 Transmittance Multispectral Imaging for Edible Oil Quality Assessment , <i>Optica Imaging and Applied Optics Congress</i> , 2020. In-house multispectral imaging system estimating coconut oil adulteration with palm oil from transmittance spectra. Skills: Optical Instrumentation, Data Acquisition, Spectrum Analysis	
📄 Transmittance Multispectral Imaging for Reheated Coconut Oil Differentiation , <i>IEEE Access</i> , 2022. Designed a cost-effective multispectral imaging system to reliably distinguish reheated oils via transmittance spectra. Skills: Optical System Design, Image Processing	

EXPERIENCE

Teaching Assistant , Purdue University, USA.	2022–2025
Assisted courses including Electrical Engineering Fundamentals I, Digital Signal Processing I, and Machine Learning. Conducted TA sessions, labs, and provided feedback on assignments, developing mentorship, communication, and organizational skills.	
Temporary Instructor , University of Peradeniya, Sri Lanka.	2020–2022
Instructed and supervised undergraduate communication labs; TA for Network Analysis, Electromagnetic Theory, DSP, Communication Systems, and Telecommunication; managed third-year lab scheduling and project supervision.	

INTEREST

Signal Processing, Machine Learning, Remote Sensing, Computational Imaging, LiDAR and RADAR

ACHIEVEMENTS

SRC JUMP 2.0 CogniSense Center Best Poster Award for the adaptive SP-LiDAR Compression Framework.	2025
Prof. E.F. Bartholameusz Endowment Award for the best undergraduate project in Engineering Mathematics.	2020
W.P. Jayasekara Prize for the best undergraduate project in the Electrical and Electronic Department.	2020